

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester V Examination, 2022-23

Paper No. CHB-505: Environmental & Nuclear Chemistry

Time: 3 Hours

Maximum Marks: 70

Instructions: Answer five questions in total, including Question 1 from each section which is compulsory. Use separate answer books for Section A and Section B.

SECTION A

(Environmental Chemistry – Marks: 35)

Question 1. Answer all parts of the following:

[5 × 2 = 10]

- (a) Differentiate between receptor and sink in environmental chemistry.
- (b) Differentiate between lithosphere and biosphere.
- (c) Discuss about the following: (i) Sludge, (ii) Van Allen Belts, and (iii) Pollution Pathways.
- (d) Explain soot and contaminant.
- (e) Discuss major regions of the atmosphere.

Question 2.

- (a) Describe various environmental segments. How can we divide the major regions of the atmosphere? State temperature variations and chemical species present. [8]
- (b) Discuss the greenhouse effect and global warming. What are its major consequences? [6]

Question 3. Account for any four of the following:

[4 × 3.5 = 14]

- (a) Minamata incident and Bhopal Gas Tragedy
- (b) Wastewater treatment of domestic effluents
- (c) Air pollution and important air quality parameters (BOD and COD)
- (d) Auto-exhaust emissions and their catalytic converters controls
- (e) Stratospheric ozone depletion chemistry

SECTION B

(Nuclear Chemistry – Marks: 35)

Question 4. Answer all parts of the following: [5 × 2 = 10]

- (a) What is the basis of using radioisotopes for the age determination of historical and archaeological organic samples (radiocarbon dating)?
- (b) What is the magic number of nucleons? Give two experimental evidences confirming the existence of magic numbers.
- (c) Differentiate between nuclear reaction cross section and geometric cross section.
- (d) Predict the modes of decay for neutron-rich and neutron-deficient nuclides.
- (e) Explain the principle of isotopic dilution analysis.

Question 5.

- (a) What are the different types of nuclear reactions? Describe radiative capture and transfer reactions in detail. [6]
- (b) Show that the kinetic energy available for exciting the target nucleus is the kinetic energy of the projectile and target in the Centre of Mass System (CMS). [6]
- (c) Write the precautions to be taken in the use of radioisotopes as tracers. [2]

Question 6.

- (a) Discuss the compound nucleus theory of nuclear reactions. Describe the Bohr model of nuclear fission. [8]
- (b) Discuss the nuclear shell model, showing magic numbers and spin-parity predictions. [6]

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